Enzyme reactions II

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Biomatematica 2

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Time constant of decay dynamics

Consider an ODE of the form

$$\begin{cases} \dfrac{dy}{dt} = -ay + b \\ y(0) = 0 \end{cases}$$

with $a, b > 0$.

We call time constant of the associated dynamics the quantity

$$\tau_y = \frac{1}{a}$$

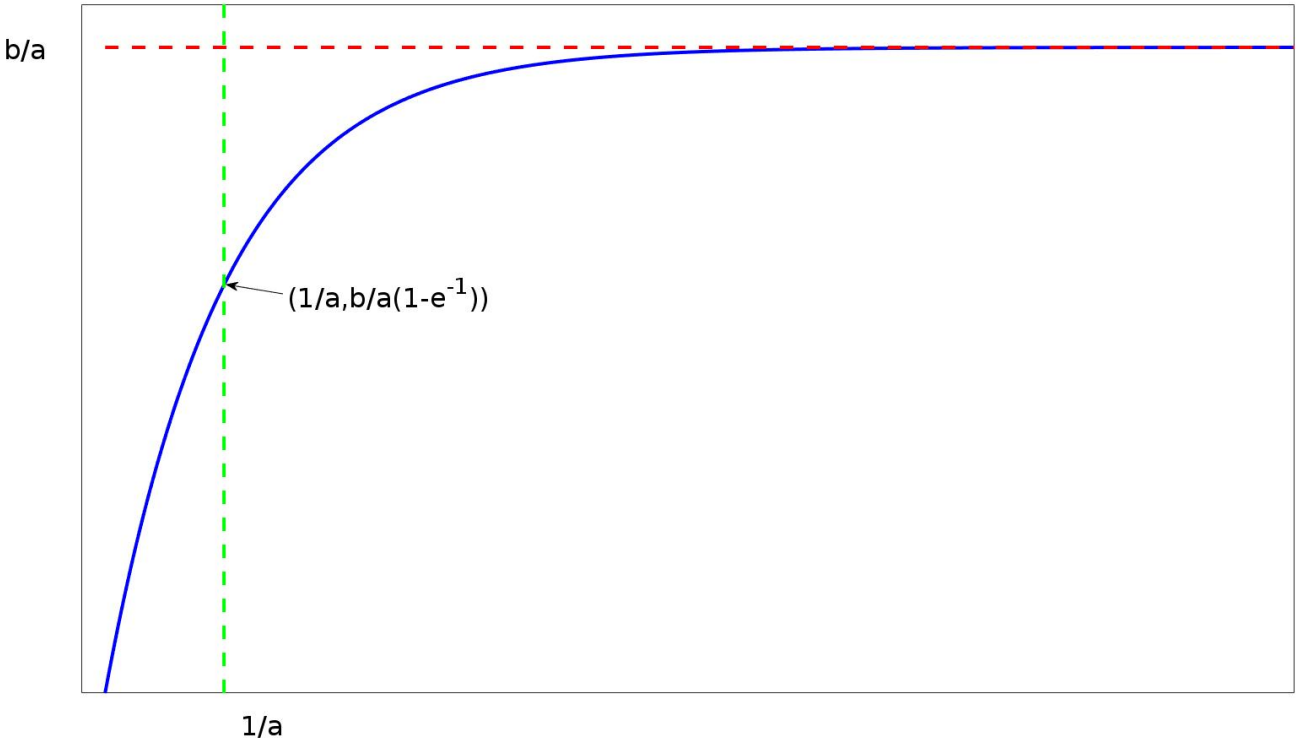
Meaning of time constant

The solution of the previous Cauchy problem is

$$y(t) = \frac{b}{a}\left(1 - e^{-at}\right).$$

Thus

$$y(\tau_y) = y\left(\frac{1}{a}\right) = \frac{b}{a}\left(1 - e^{-1}\right) \approx \frac{2}{3}\frac{b}{a}.$$



Time constants of the transient state

During the transient state, the dynamics is mainly determined by equation

$$\frac{dc}{dt} = -(k_{-1} + k_2 + k_1 s_0)c + k_1 e_0 s_0.$$

Since $k_{-1} + k_2 << k_1 s_0$, it follows $k_{-1} + k_2 + k_1 s_0 \approx k_1 s_0$.

Therefore the time constant of the transient state is given by

$$\tau_r = \frac{1}{k_1 s_0}.$$

Time constants of the quasi-steady state

During the quasi-steady state, the dynamics is mainly determined by equation

$$\frac{ds}{dt} = -k_1 e_0 s + (k_1 s + k_{-1})c.$$

Therefore the time constant of the quasi-steady state is given by

$$\tau_p = \frac{1}{k_1 e_0}.$$

Remark: the transient state is much faster than the quasi-steady state, because

$$e_0 << s_0 \qquad \Rightarrow \tau_r << \tau_p.$$

Uniform approximation

We now construct a uniform approximation of the solution of two-stage kinetics system, holding not only during the *quasi-steady state* but also during the *transient state*.

Its construction is based of the following steps:

1. Derivation of the dimensionaless system
2. Construction of outer solution
3. Construction of inner solution
4. Matching

Dimensionless system

Let $\varepsilon := \frac{e_0}{s_0}$ and $\tau = \frac{t}{\tau_p} = k_1 e_0 t$, which implies $\frac{dt}{d\tau} = \frac{1}{k_1 e_0}$ and consider

$$\textit{dimensionless } s: \quad u(\tau) = \frac{s(t)}{s_0} \qquad \Rightarrow s(t) = s_0 u(\tau)$$

$$\textit{dimensionless } c: \quad v(\tau) = \frac{c(t)}{e_0} \qquad \Rightarrow c(t) = e_0 v(\tau).$$

By deriving u and v, we obtain:

$$\frac{du}{d\tau} = \frac{1}{s_0}\frac{ds}{dt}\frac{dt}{d\tau} = \frac{1}{k_1 e_0 s_0}\frac{ds}{dt} \qquad \Rightarrow \frac{ds}{dt} = k_1 e_0 s_0 \frac{du}{d\tau}$$

$$\frac{dv}{d\tau} = \frac{1}{e_0}\frac{dc}{dt}\frac{dt}{d\tau} = \frac{1}{k_1 e_0^2}\frac{dc}{dt} \qquad \Rightarrow \frac{dc}{dt} = k_1 e_0^2 \frac{dv}{d\tau}$$

Dimensionless system II

By substituting in the system the expressions for $\frac{ds}{dt}$ and $\frac{dc}{dt}$, we get:

$$\begin{cases} \dfrac{du}{d\tau} = -u(\tau) + \left(u(\tau) + \dfrac{k_{-1}}{k_1 s_0}\right) v(\tau) & u(0) = \frac{s(0)}{s_0} = 1 \\ \dfrac{e_0}{s_0}\dfrac{dv}{d\tau} = -\left(\dfrac{k_{-1}+k_2}{k_1 s_0} + u(\tau)\right) v(\tau) + u(\tau) & v(0) = \frac{c(0)}{e_0} = 0 \end{cases}$$

Since $\varepsilon := \frac{e_0}{s_0}$ and denoting by $\alpha := \frac{k_{-1}}{k_1 s_0}$, $K := \frac{k_{-1}+k_2}{k_1 s_0}$, the dimensionless system reads:

$$\begin{cases} \dfrac{du}{d\tau} = -u + (u+\alpha)v & u(0) = 1 \\ \varepsilon\dfrac{dv}{d\tau} = -(K+u)v + u & v(0) = 0 \end{cases} \tag{1}$$

Dimensionless system III

System (1) is a singular perturbation problem, since letting $\epsilon = 0$, the system becomes a differential-algebraic equations system instead of an ODE system.

Finding a uniform approximation for (u, v) means finding $(\tilde{u}, \tilde{v})$ such that

$$\begin{cases} |u(\tau) - \tilde{u}(\tau)| = O(\varepsilon) \\ |v(\tau) - \tilde{v}(\tau)| = O(\varepsilon) \end{cases} \qquad \forall \tau \tag{2}$$

Outer solution

Let us seek solutions u and v of the form:

$$u(\tau) = \sum_{n=0}^{+\infty} \varepsilon^n u_n(\tau)$$

$$v(\tau) = \sum_{n=0}^{+\infty} \varepsilon^n v_n(\tau)$$

Substituting such expressions in system (1), we obtain for $n = 0$:

$$\begin{cases} \dfrac{du_0}{d\tau} = -u_0 + (u_0 + \alpha)v_0 \qquad u_0(0) = 1 \\ 0 = -(K + u_0)v_0 + u_0 \end{cases}$$

Outer solution II

We observe that v_0 does not satisfy the initial condition because

$$v_0(0) = \frac{u_0(0)}{K + u_0(0)} = \frac{1}{1+K} \neq 0.$$

Hence (2) does not hold, $\Rightarrow$ (u_0, v_0) is not the uniform solution.

However, (u_0, v_0) is the dimensionless quasi-steady state approximation, i.e.

$$\begin{cases} |u(\tau) - u_0(\tau)| = O(\varepsilon) \\ |v(\tau) - v_0(\tau)| = O(\varepsilon) \end{cases} \qquad \underline{\forall \tau > \tau_0 > 0}$$

(u_0, v_0) is called outer solution, because it is a good approximation of (u, v) only outside the transient state.

Inner solution

We construct now an approximation of (u, v) that holds inside the transient state (layer).

Introduce the new variable $\sigma = \frac{\tau}{\varepsilon} = k_1 s_0 t$, hence

$$U(\sigma) = u(\tau) \quad \Rightarrow \frac{dU}{d\sigma} = \frac{du}{d\tau}\frac{d\tau}{d\sigma} = \varepsilon\frac{du}{d\tau} \quad \Rightarrow \frac{du}{d\tau} = \frac{1}{\varepsilon}\frac{dU}{d\sigma}$$

$$V(\sigma) = v(\tau) \quad \Rightarrow \frac{dV}{d\sigma} = \frac{dv}{d\tau}\frac{d\tau}{d\sigma} = \varepsilon\frac{dv}{d\tau} \quad \Rightarrow \frac{dv}{d\tau} = \frac{1}{\varepsilon}\frac{dV}{d\sigma}$$

Substituting the expressions of $\frac{du}{d\tau}$ and $\frac{dv}{d\tau}$ in system (1) and multiplying times ε the first equation, we obtain:

$$\begin{cases} \dfrac{dU}{d\sigma} = \varepsilon(-U + (U + \alpha)V) & U(0) = 1 \\ \dfrac{dV}{d\sigma} = -(K + U)V + U & V(0) = 0 \end{cases} \tag{3}$$

Inner solution II

The previous system represents a *regular perturbation problem*, because when $\varepsilon = 0$ the system remains an ODE system.

We seek now U and V of the form:

$$\begin{cases} U(\sigma) = \displaystyle\sum_{n=0}^{+\infty} \varepsilon^n U_n(\sigma), \\ V(\sigma) = \displaystyle\sum_{n=0}^{+\infty} \varepsilon^n V_n(\sigma) \end{cases}$$

Inner solution III

Substituting such expressions in system (1) we obtain for $n = 0$:

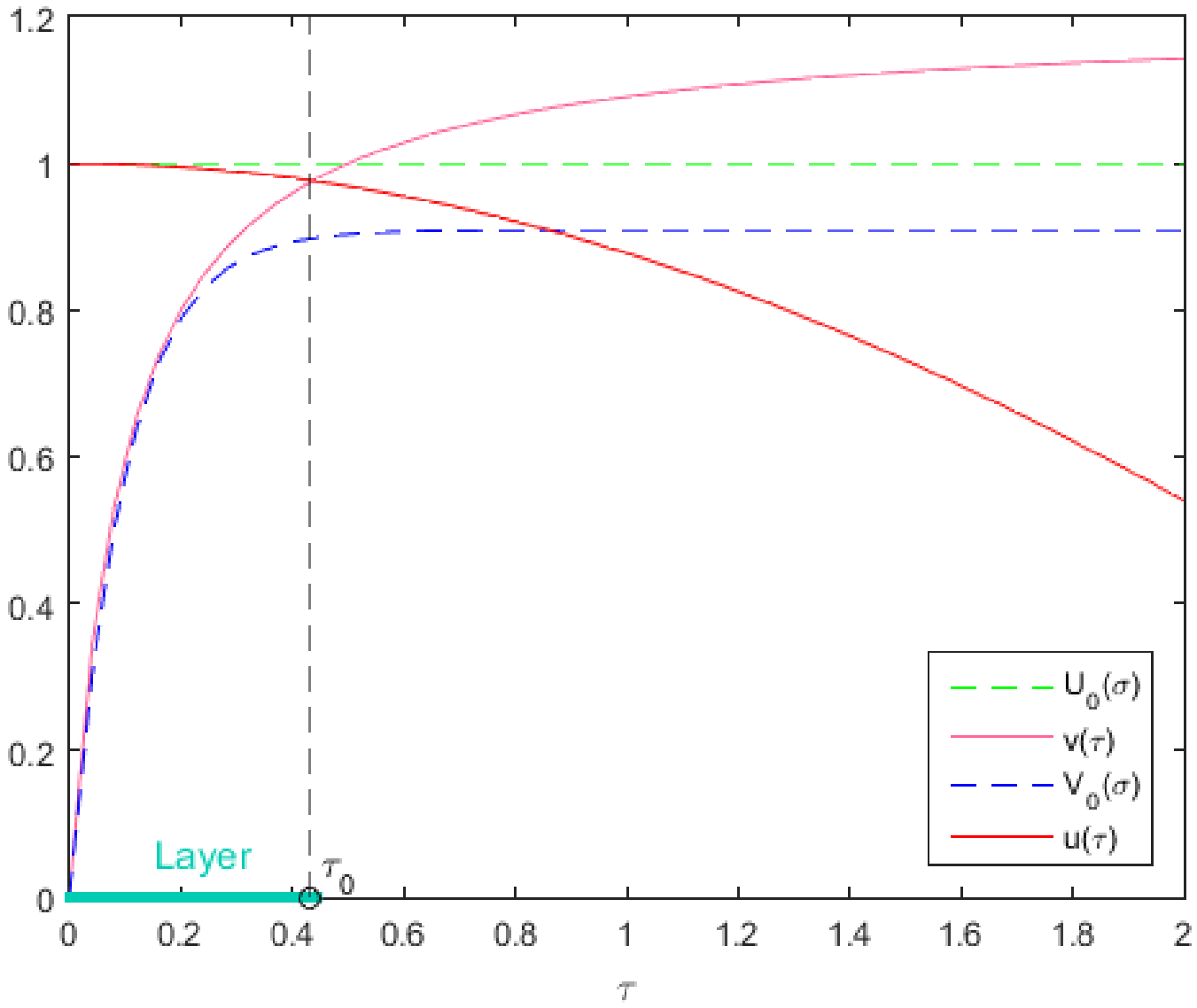
$$\begin{cases} \dfrac{dU_0}{d\sigma} = 0 & U_0(0) = 1 \\ \dfrac{dV_0}{d\sigma} = -(K + U_0)V_0 + U_0 & V_0(0) = 0 \end{cases}$$

This implies

$$\begin{cases} U_0(\sigma) = 1 \\ V_0(\sigma) = \frac{1-e^{-(K+1)\sigma}}{K+1}. \end{cases}$$

(U_0, V_0) is called inner solution, because it is a good approximation of (u, v) only inside the transient state.

Inner solution in picture



Matching

The matching conditions to be verified at the layer interface reads:

$$\lim_{\sigma \longrightarrow +\infty} U_0(\sigma) = \lim_{\tau \longrightarrow 0} u_0(\tau), \qquad \lim_{\sigma \longrightarrow +\infty} V_0(\sigma) = \lim_{\tau \longrightarrow 0} v_0(\tau).$$

Indeed

$$\begin{cases} \lim_{\sigma \longrightarrow +\infty} U_0(\sigma) = \lim_{\sigma \longrightarrow +\infty} 1 = 1 \\ \lim_{\tau \longrightarrow 0} u_0(\tau) = u_0(0) = 1 \end{cases} =: L_u$$

$$\begin{cases} \lim_{\sigma \longrightarrow +\infty} V_0(\sigma) = \lim_{\sigma \longrightarrow +\infty} \frac{1}{K+1}(1 - e^{-(K+1)\sigma}) = \frac{1}{K+1} \\ \lim_{\tau \longrightarrow 0} v_0(\tau) = \lim_{\tau \longrightarrow 0} \frac{u_0(\tau)}{K + u_0(\tau)} = \frac{1}{K+1} \end{cases} =: L_v$$

Matching II

The uniform approximation is given by:

$$\begin{cases} \tilde{u}(\tau) = u_0(\tau) + U_0(\tau) - L_u = u_0(\tau) \\ \tilde{v}(\tau) = v_0(\tau) + V_0(\tau) - L_v = \dfrac{u_0(\tau)}{K + u_0(\tau)} - \dfrac{e^{-(K+1)\frac{\tau}{\varepsilon}}}{K+1} \end{cases}$$

It can be proved that $(\tilde{u}(\tau), \tilde{v}(\tau))$ satisfies

$$\begin{cases} |u(\tau) - \tilde{u}(\tau)| = O(\varepsilon) \\ |v(\tau) - \tilde{v}(\tau)| = O(\varepsilon) \end{cases} \qquad \forall \tau$$

Uniform approximation in dimensionless form

Summarizing, to compute the uniform approximation, we have to solve the following system:

$$\begin{cases} \dfrac{d\tilde{u}}{d\tau} = -\tilde{u} + (\tilde{u} + \alpha)\dfrac{\tilde{u}}{K + \tilde{u}} = \dfrac{(\alpha - K)\tilde{u}}{K + \tilde{u}} \qquad \tilde{u}(0) = 1 \\ \tilde{v}(\tau) = \dfrac{\tilde{u}(\tau)}{K + \tilde{u}(\tau)} - \dfrac{e^{-(K+1)\frac{\tau}{\varepsilon}}}{K + 1} \end{cases} \tag{4}$$

Uniform approximation in dimensional form

To compute the uniform approximation in dimensional form $(\tilde{s}, \tilde{c})$, we have to solve the following system:

$$\begin{cases} \dfrac{d\tilde{s}(t)}{dt} = -\dfrac{k_2 e_0 \tilde{s}}{k_m + \tilde{s}} \\ \tilde{c}(t) = \dfrac{e_0 \tilde{s}}{k_m + \tilde{s}} - \dfrac{e_0 s_0}{k_m + s_0} e^{-\left(\frac{k_m}{s_0}+1\right) k_1 s_0 t} \end{cases} \qquad \tilde{s}(0) = s_0 .$$

Esercizio

Dall'adimensionamento di un modello di reazione chimica si ottiene il seguente problema di perturbazione singolare:

$$\begin{cases} \dfrac{du}{dt} = uv - 2 & u(0) = 1 \\ \epsilon \dfrac{dv}{dt} = -v + 3 & v(0) = 0. \end{cases}$$

Calcolarne l'approssimazione uniforme $(\widetilde{u}(t), \widetilde{v}(t))$.